

Yuan-Sheng (Jan) Hsu — M.D., Ph.D.

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Research Summary

Physician-scientist with research interests in bone and mineral health and clinical pharmacoepidemiology. Current work focuses on polygenic risk scores for osteoporotic fracture prediction in a Taiwanese DXA-linked biobank cohort, real-world evidence studies of skeletal disease using federated networks, and causal-inference methods (target trial emulation, propensity score, E-value sensitivity). Earlier doctoral training in basic biomedical research provides a foundation for evaluating mechanistic, translational, and epidemiologic evidence across scales.

Research Interests

Bone and mineral metabolism | Polygenic risk scores and fracture prediction | Real-world evidence and pharmacoepidemiology | Causal inference for observational data

Education

China Medical University

M.D., Ph.D. in Medicine

Taichung, Taiwan

–2024

Ph.D. advisor: Academician Wen-Hwa Lee, Genomics Research Center, Academia Sinica

Dissertation: Differential effects of glucose and N-acetylglucosamine on genome instability

Honors and Awards

2026: Best Abstract Presentation Award (EGPT session), 41st Annual European Association of Urology (EAU) Congress, London

2023: Competitive Doctoral Dissertation Award, TSC Thesis Competition, Taiwan

2021: Certificate of Outstanding Poster, International Conference on Advanced Biomedical Sciences (ICABS), Taichung

Academic and Clinical Positions

Taichung Veterans General Hospital

Physician (PGY)

Taichung, Taiwan

Mar 2025–Present

- Lead an independent research program in bone and mineral health, mentored by Dr. Kun-Hui Chen (Chairman, Department of Orthopedic Surgery)
- Develop and validate polygenic risk scores derived from heel eBMD GWAS for major osteoporotic fracture prediction in a Taiwanese DXA-linked cohort, using Fine-Gray competing risk models, restricted cubic splines, and BMD-stratified subgroup analysis
- Conduct multi-departmental real-world evidence studies using the TriNetX federated network and institutional cohorts, applying target trial emulation, propensity score methods, and E-value sensitivity analysis

Ministry of National Defense

Substitute Military Service

Hualien, Taiwan

Aug 2024–Mar 2025

China Medical University Hospital

Clinical Data Analyst (concurrent with clerkship)

Taichung, Taiwan

Sep 2022–Aug 2024

- Designed and queried a hospital-scale data warehouse (~400,000 patient samples) using SQL and ICD diagnostic codes to define oncologic and matched-control cohorts
- Executed genome-wide association analyses on Linux using PLINK; identified significant loci and generated LocusZoom visualizations in a 1:5 matched case-control study of breast cancer

Genomics Research Center, Academia Sinica

Doctoral Researcher

Taipei, Taiwan

Mar 2018–Jul 2022

- Investigated mechanisms by which high-sugar diets induce pancreatic DNA damage through O-GlcNAcylation and nucleotide-metabolism disruption in the Wen-Hwa Lee laboratory
- Designed animal model studies comparing DNA damage across sugar types and dosages; performed LC-MS metabolomics for dNTP pool quantification
- Developed 96-well high-content imaging assays for DNA damage and protein glycosylation; conducted multiplex IHC with RNA SCOPE on precancerous pancreatic lesions

Peer-Reviewed Publications

First Author: Hsu YS, Tseng SC, Chao TF, Chen KH. Osteoporosis risk in alpha and beta thalassemia: an age- and sex-specific retrospective cohort study. *Archives of Osteoporosis*. 2026;21:83. doi:10.1007/s11657-026-01711-y

Hsu JYS, Li JR, Chen CS, Yang CJ. Association of phosphodiesterase type 5 inhibitor use with cardiovascular and renal outcomes in male kidney transplant recipients with benign prostatic hyperplasia or erectile dysfunction. *International Urology and Nephrology*. 2026. doi:10.1007/s11255-026-05192-3

Hsu YS, Wu PJ, Jeng YM, Hu CM, Lee WH. Differential effects of glucose and N-acetylglucosamine on genome instability. *American Journal of Cancer Research*. 2022;12(4):1556–1576.

Hsu YS, Yong SB. Correspondence to “Early infancy dysbiosis in food protein-induced enterocolitis syndrome: a prospective cohort study”. *Allergy*. 2024;79(2):540–541.

Co-author: Liu YH, Hu CM, **Hsu YS**, Lee WH. Interplays of glucose metabolism and KRAS mutation in pancreatic ductal adenocarcinoma. *Cell Death & Disease*. 2022;13(9):817.

Manuscripts Under Review

First Author: Hsu YS, et al. A polygenic risk score improves fracture prediction beyond bone density in peri-menopausal women in Taiwan: a TPML cohort study.

First Author: Hsu YS, et al. PSA density predicts biochemical recurrence after radical prostatectomy in low- and intermediate-risk prostate cancer: an androgen-deprivation-naïve cohort study with landmark analysis.

First Author: Hsu YS, et al. Preoperative vitamin D deficiency and 10-year hypocalcemia, arrhythmia, and depression after total thyroidectomy.

Conference Presentations

2026: Hsu YS, et al. A polygenic risk score improves fracture prediction beyond bone density in peri-menopausal women in Taiwan: a TPML cohort study. *ASBMR 2026 Annual Meeting* (Plenary Poster Session), Boston, MA, USA, October 2026.

2026: Hsu JYS, Li JR, Yang CJ. Tadalafil improves cardiovascular and renal outcomes in kidney transplant recipients with benign prostate hyperplasia or erectile dysfunction. *41st Annual EAU Congress*, London, UK, March 2026.

IRB-Approved Research Protocols

Studies co-designed and conducted as study investigator. Institutional principal investigator listed per local regulation.

CE25895A: Polygenic risk scores and opportunistic bone screening for identifying individuals at high risk of osteoporosis. PI: K.-H. Chen. (Ongoing)

CE26402C: Pharmacologic and clinical risk factors for complications after joint arthroplasty. PI: K.-H. Chen. (Ongoing)

CE25652C: Thalassemia subtypes and sex differences in osteoporosis risk: a real-world data study. PI: K.-H. Chen. (Completed; published)

CE26517B: Clinical characteristics and polygenic scores for predicting biochemical recurrence after radical prostatectomy in a Taiwanese Han cohort. PI: J.-R. Li. (Ongoing)

CE26634C: Effect of 5 α -reductase inhibitors on outcomes of prostatic disease treatment. PI: J.-R. Li. (Ongoing)

CE26369B: Preoperative vitamin D status for predicting hypocalcemia and chronic disease after total thyroidectomy: a multi-institution real-world cohort. PI: W.-H. Chen. (Ongoing)

CE26584C: Polygenic scores for ischemic stroke risk prediction in atrial fibrillation: a TPML retrospective cohort. PI:

Y.-S. Huang. (Ongoing)

CE26327C: Major bleeding risk during oral anticoagulation in atrial fibrillation: validation of combined polygenic and clinical risk scores. PI: Y.-S. Huang. (Ongoing)

Research Skills

Epidemiology & Statistics: Survival analysis (Cox, Fine-Gray competing risk), polygenic risk score derivation and validation (PRS-CS, PLINK, LDpred2), target trial emulation, propensity score methods, restricted cubic splines, E-value sensitivity analysis

Programming & Data: R (tidyverse, survival, rms), Python (pandas, scikit-learn, PyTorch), SQL, Linux, Git, Jupyter

Genomics & Bioinformatics: PLINK, GWAS pipelines, LocusZoom, PGS Catalog integration

Biomedical Research: Western blot, IP, IF, ELISA, LC-MS metabolomics, high-content imaging, multiplex IHC with RNA SCOPE, spinning-disk confocal microscopy, animal model design

Research Databases: TriNetX federated network, Taiwan Precision Medicine Initiative (TPMI), hospital data-warehouse design

Languages

Mandarin Chinese (native) | English (fluent) | Japanese (JLPT N1) | German (Goethe B1)